

Why Borrowers Fail to Refinance: Awareness, Mental Cost, and Time Discounting

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Outline

- 1 Motivation
- 2 Institutional Background and Data
- 3 Structural Model
- 4 Estimation and Results

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Motivation: Why under-refinancing matters

- Many eligible borrowers **do not refinance** despite large savings opportunities. (Keys–Pope, 2016)
- HARP expected **4–5M** take-ups; by 2011, **<1M** had participated → repeated rule changes + awareness campaign. (FHFA OIG, 2013)
- **Macroeconomic channel:** fewer refis = weaker pass-through from rate cuts to household cash flow and spending. (Amromin–Bhutta–Keys, 2020)
- Missed savings were **material** (media est. $\sim$ \$13B/yr during low-rate years). (NerdWallet, 2016; JCHS, 2018)

Empirical Puzzle

- Large, well-documented **failure to refinance** despite substantial gains (Keys–Pope, 2016; Andersen et al., 2020).
- Borrower-reported frictions: **awareness**, **paperwork**, and **process difficulty**. (Fannie Mae NHS 2013)
- Three candidate channels: **inattention**, **mental cost**, **impatience**.
- This paper: a dynamic model that separates the channels and quantifies each.

Related Literature and Our Contribution

Agarwal, Driscoll & Laibson (2013, JMCB)

- Closed-form **threshold** for optimal refinancing — the clear **benchmark incentive** measure.
- Assumes **full attention**, calibrated discounting, no mental cost.

Andersen et al. (2020, AER)

- Decompose inaction into **psychological cost** vs. **inattention** (Danish administrative data).
- Relative to their design, we move:
 - threshold rule → full **dynamic optimization** (option value, program deadline);
 - i.i.d. inattention → **persistent awareness** diffusing with local exposure: identified by an *exclusion restriction*, not functional form;
 - frictionless Denmark → U.S. **program setting**, where eligibility and outreach design matter.

⇒ **Policy counterfactuals on the levers FHFA actually pulled**: universal awareness, fee design.

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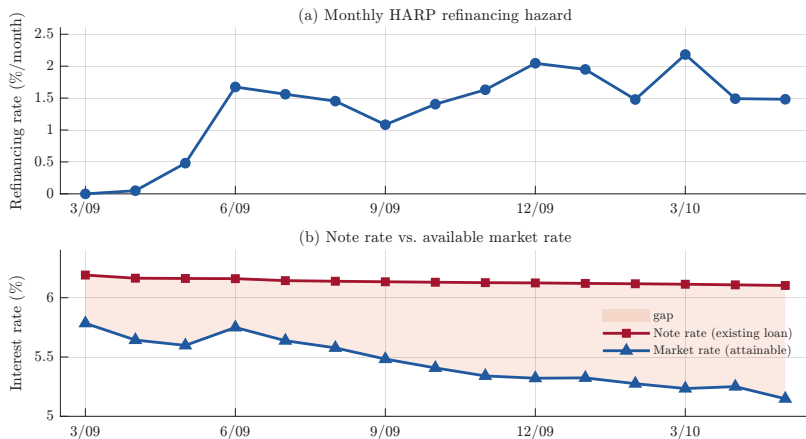
The HARP Program: targeted underwater borrowers

- **HARP 1.0**: announced March 2009 by **FHFA**, implemented through Fannie Mae and Freddie Mac (under the Making Home Affordable umbrella).
 - Loan-to-value ratio (LTV) $\in (80\%, 105\%]$; cap raised to 125% in July 2009.
 - Loan **owned or guaranteed by Fannie/Freddie** and originated **on or before 31 May 2009**.
 - Borrower **current** on payments — no 30-day delinquency in the prior 12 months.
- **HARP 2.0** (announced October 2011): major expansion.
 - Removed the LTV cap for **fixed-rate** loans (ARMs kept a 105% ceiling).
 - Reduced lender representations and warranties; simplified appraisal and underwriting.
- Deadline extended repeatedly; program closed December 2018.

Loan Data Overview

- Freddie Mac Single-Family Loan-Level Dataset (1999–2023).
- Sample: 110,922 HARP-eligible loans, 4.2M loan-months.
- Controls: credit score, LTV, origination year, first-time homeowner.

Stylized Facts



- Take-up is **slow**: 0.00%, 0.05%, 0.48% at launch, then $\sim 1.5\%/month$.
- Meanwhile the gap **widens** — market rate $-64bp$, note rates flat, so the gap grows $0.41\% \rightarrow 0.96\%$ ($2.4\times$).

Survey evidence on the cost channel

Fannie Mae National Housing Survey (2013Q1): mortgage holders who had **not refinanced** ($n = 609$), share citing each barrier:

Barrier	% citing
Wouldn't reduce payments enough	52%
Closing costs too high	45%
Don't want to lengthen loan term	43%
Not sure what to trust with lenders	32%
Process too complicated	31%
Would need to put money down	29%

- The **monetary** cost is real (45%) and observable: we calibrate $\kappa(B)$.
- Beyond it, **non-pecuniary** frictions — process complexity (31%) and distrust of lenders (32%) — are what the *mental cost* captures.
- “Would need to put money down” (29%) is a *misperception* under HARP: evidence of an *information friction*.

Source: Huang, “What Motivates Borrowers to Refinance?”, Fannie Mae National Housing Survey, Jan–Mar 2013.

External benchmark: discount factors

- **Blevins, Shi, Haurin & Moulton (2020, IER)**: DDC of reverse-mortgage (HECM) borrowers, 2007–2014. The discount factor is *identified* and *estimated*:

$$\hat{b} = 0.898 \text{ (annual)}, \quad 95\% \text{ CI } [0.71, 1.00]$$

- Experimental and field estimates for comparable populations range **0.67–0.90** (Trostel–Taylor 2001; Harrison–Lau–Williams 2002; Read–Read 2004).
- **Frederick–Loewenstein–O’Donoghue (2002, JEL)** survey 43 studies: median implied rate $\approx 35\%$ /year, with enormous dispersion. Critically, the implied rate **falls steeply with horizon**:

Horizon	1 month	1 year	10 years
Implied annual rate	345%	120%	19%
Implied β_{ann}	0.22	0.45	0.84

- **Takeaway:** estimated β well below 1 is the *norm*, and the shorter the horizon the lower the factor.

Survey evidence on the attention channel

- FHFA (2013) survey: only 34–46% of eligible non-refinancers had ever contacted a lender.
- FHFA cited inattention as a key barrier
- FHFA launched a nationwide awareness campaign in September 2013.
 - FHFA kicked off a nationwide public-education push
 - Unveiled HARP.gov,
 - Partnered with mortgage companies and HGTV personality Mike Aubrey to reach eligible borrowers via TV/radio/online.

The attention shifter: local HARP exposure

X_t = **local refinancing rate** from program launch through $t - 1$: exposure to HARP activity in the borrower's market.

Refi hazard on local exposure		
	Coef.	t
No fixed effects	0.98	11.0
+ month FE	1.06	9.0
+ month FE + DMA FE	0.89	9.3
+ both FE + current local refi rate	0.40	8.5

- Local exposure, enters the *awareness process only*, excluded from utility $\Rightarrow$ exclusion restriction.

Summary: Channels for inaction

- 1 **Discounting:** Borrowers may apply a much lower discount rate to future savings than benchmark calibrations assume.
- 2 **Mental cost:** Non-pecuniary frictions (effort, paperwork, uncertainty) plausibly comparable in size to the closing fee.
- 3 **Inattention:** A non-trivial share of eligible borrowers may not actively consider refinancing in any given period.

These three channels motivate the dynamic discrete-choice model below, which lets us estimate each separately and assess their relative contributions to under-refinancing.

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Model simplification assumptions

- Many changes occurred during the course of HARP.
 - We restrict the structural sample to the first program window, **March 2009 – May 2010** ($T = 15$ months), during which borrowers expected the program to expire in June 2010.
 - Focus on borrowers' behaviors, without complications from lenders.
- Refinance assumptions
 - Do not model refinance into different terms or cash-out refinance.
 - Focus exclusively on refinancing into a new 30-year fixed-rate loan, which accounts for over half of observed refinance activity.
 - Given the one-year program duration, only refinance once (terminating action).

Dynamic Decision Process

Finite horizon: $t = 1, \dots, T$ (HARP expiration)

Stage 1: Awareness

$$\Pr(a_t = 1 | X_t, a_{t-1}) = \mathbf{1}[a_{t-1} = 0] \frac{e^{X_t \tau}}{1 + e^{X_t \tau}} + \mathbf{1}[a_{t-1} = 1].$$

- Unaware borrowers become aware with probability $\phi(X_t)$, driven by the attention shifter.
- Awareness is **absorbing**: once aware, borrowers stay aware. (Forgetting δ is not separately identified in a single-market rollout — it reveals itself only when exposure pauses, which never occurs in our window; we impose $\delta = 0$.)
- If $a_t = 0$, refinancing is infeasible.

Stage 2: Refinance decision (if aware)

$$d_t \in \{0 \text{ (no refi)}, 1 \text{ (refi)}\}.$$

Economic Environment

State variables:

$$S_t = (P_t, B_t, age_t, m_o, Z), \quad r_t$$

- B_t : outstanding balance
- P_t : house value
- m_o : original scheduled payment
- age_t : loan age
- Z : borrower characteristics (first homeowner, credit score)
- r_t : feasible refi rate

Refinancing is terminating. After $d_t = 1$, no future refinancing allowed.

Value of Not Refinancing ($d_t = 0$)

Flow utility:

$$\tilde{u}_0 = u_0(S_t) + \epsilon_{0t}.$$

Choice-specific value:

$$v_{0t}(S_t, r_t) = u_0(S_t) + \beta P_m(S_t) E[V_{t+1}(S_{t+1}, r_{t+1}) | S_t, r_t, d_t = 0].$$

- Must continue paying to keep the **option to refinance**.
- Option value embedded in V_{t+1} .

Value of Refinancing ($d_t = 1$)

Flow utility:

$$\tilde{u}_1 = -\kappa(B_t) - \rho(Z) + \epsilon_{1t}.$$

The closing fee is **calibrated**, not estimated:

$$\kappa(B_t) = \underbrace{0.01 B_t + \$2,000}_{1\% \text{ of balance} + \text{flat}}$$

Choice-specific value:

$$v_1(S_t, r_t) = u_1(S_t) + \beta E[L(S_{t+1}) | S_t, r_t, d_t = 1].$$

Continuation without refi option:

$$L(S_t) = u_0(S_t) + \beta P_m(S_t) E[L(S_{t+1}) | S_t, d_t = 0].$$

- After refinancing $\rightarrow$ no more option value.
- Future value purely amortization value.

Optimal Refinancing Decision

Refinance if

$$v_1(S_t, r_t) + \epsilon_{1t} \geq v_0(S_t, r_t) + \epsilon_{0t}.$$

Logit CCP (EV1 shocks):

$$\Pr(d_t = 1 | a_t = 1, S_t, r_t) = \frac{e^{v_1(S_t, r_t)}}{e^{v_1(S_t, r_t)} + e^{v_0(S_t, r_t)}}.$$

If $a_t = 0$, $\Pr(d_t = 1) = 0$.

Terminal Period T

No future option value.

$$v_{0T} = u_0(S_T) + \beta P_m(S_T) E[L(S_{T+1}) | d_T = 0]$$

$$v_{1T} = u_1(S_T) + \beta E[L(S_{T+1}) | d_T = 1]$$

Static logit comparison.

Backward induction solves $t = T, \dots, 1$.

Identification

- If attention is observed:
 - Finite horizon.
 - Refinancing as terminal action, following identification strategy in Blevins et al. (2020, IER)
- Unobserved attention
 - Attention shifter (lagged local refinancing rate) enters awareness transition but not utility: exclusion restriction.
 - Panel data structure helps identify the attention process over time.

Empirical Specification: Primitives

1. Flow Utility (No Refi relative to the base utility of refi)

$$u_0(S_t; \theta_u) = [1 \quad m_o \quad \text{equity}_t \quad \text{FirstHome} \quad \ln(\text{CreditScore})] \theta_u$$

2. Mental (Switching) Cost: Depends only on a borrower-level fixed characteristic

$$\rho(Z; \theta_\rho) = \theta_0 + \theta_\rho \text{FirstHome}$$

Empirical Specification: Awareness

3. Awareness Process

Initial awareness — tied to the onset process, not a free parameter:

$$\Pr(a_{i1} = 1) = \phi(X_1; \theta_a) \big|_{X_1=0} = \frac{e^{\phi_c}}{1 + e^{\phi_c}} \equiv \alpha_1$$

Transition (if previously inattentive):

$$\phi(X_t; \theta_a) = \frac{\exp(X_t' \theta_a)}{1 + \exp(X_t' \theta_a)},$$

Awareness is *absorbing*: $\Pr(a_t = 1 \mid a_{t-1} = 1) = 1$.

X_t : lagged DMA-level refinancing rate

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Structural estimation sample

Sample: NYC DMA, 8,896 HARP-eligible borrowers, 99,595 loan-months; observations March 2009 – May 2010, with the announced June-2010 expiration as the decision horizon.

Borrowers enter the sample as they first satisfy eligibility (LTV band, delinquency screens), month by month.

Why the first program window:

- A single, stable time window: HARP 1.0 only, before the late-2011 HARP 2.0 overhaul changed eligibility and lender incentives mid-game.
- A well-defined **finite horizon**: borrowers faced the announced June-2010 expiration, giving the dynamic program its terminal period.
- Refinance-once (terminating action) is innocuous over a 15-month window.

Why New York:

- The most refinancing events among the major markets
- 26 zip3 sub-markets with widely different exposure paths $\Rightarrow$ **within-market variation** in the awareness shifter.

Estimation Results

		Estimate	Std. Err.
<i>Preferences</i>			
Discount factor (monthly)	β	0.8995	0.0068
<i>Cost frictions</i> (dollars)			
Mental cost, universal	ρ_0	\$4,576	\$162
Mental cost, FirstHome	ρ_{FH}	-\$15	\$119
Closing fee	κ_B	\$4,707	calibrated
<i>Attention</i>			
Initial aware share	α_1	0.0823	0.0153
Onset intercept	ϕ_c	-2.4112	0.2020
Onset slope	ϕ_x	11.2269	3.0496

Counterfactual Analyses

Outcome: probability a borrower refinances at some point over the 15-month window.

Each counterfactual re-solves the model

Each factor is moved **alone**, others held at their estimates.

Counterfactual Results: Each Friction Alone

Scenario	Refi rate	Δ (pp)	Δ (%)
Baseline (estimated model)	14.55%	—	—
Full attention ($\alpha_t \equiv 1$)	16.00%	+1.45	+9.9%
No mental cost ($\rho_0 = 0$)	57.01%	+42.45	+291.7%
No closing fee ($\kappa_B = 0$)	67.12%	+52.57	+361.2%
Patient ($\beta_{\text{ann}} = 0.90$)	56.50%	+41.95	+288.2%
All frictions removed	90.34%	+75.79	+520.7%

Conclusion

Facts. Substantial failure to refinance despite large potential savings, with borrowers citing awareness and process frictions

Counterfactuals.

- **Two costs, one felt and one paid: they are the same size.** The mental cost of refinancing is worth about what the closing fee costs. Removing either roughly **quadruples** take-up.
- **Impatience is a third barrier of the same magnitude.** Making borrowers patient does as much as abolishing the mental cost entirely: discounting is a *channel*
- **Attention is different in kind.** Universal awareness barely moves the level

Policy Takeaway. Fee reduction and process simplification act on barriers of equal size, and each is worth far more than an awareness campaign.

Questions?